

Ex 9

Aim

To create an interactive web-based form that validates user input (Email and Phone Number) in real-time before submission, ensuring data integrity and providing immediate user feedback.

Procedure

The development was carried out in four distinct phases:

1. Structure (HTML):

- Developed an index.html file containing a form with unique IDs (email, phone).
- Included empty `<span>` tags with the class `error` to serve as placeholders for dynamic error messages.
- Linked the **Validator.js CDN** and a custom script.js file.

2. Styling (CSS):

- Used Flexbox in style.css to center the form on the screen.
- Styled the form container with a white background, padding, and a subtle box shadow for a modern look.
- Defined an `.error` class to display validation messages in **red** with a smaller font size.

3. Logic (JavaScript):

- Used `document.getElementById` to capture form and input elements.
- Added a submit event listener that uses `e.preventDefault()` to stop the page from refreshing on submission.
- Utilized the `validator.isEmail()` and `validator.isMobilePhone(phone, 'en-IN')` methods to check the validity of the inputs.

4. User Feedback:

- Implemented conditional logic: if inputs are invalid, the corresponding error message is injected into the DOM.
- If valid, a success alert is triggered, and the data is logged to the console.

```

JS script.js X
JS script.js > ...
1 document.getElementById("myForm").addEventListener("submit", function (e) {
2     e.preventDefault();
3
4     let email = document.getElementById("email").value;
5     let phone = document.getElementById("phone").value;
6
7     let emailError = document.getElementById("emailError");
8     let phoneError = document.getElementById("phoneError");
9
10    // Clear previous errors
11    emailError.textContent = "";
12    phoneError.textContent = "";
13
14    let isValid = true;
15
16    // Validate Email
17    if (!validator.isEmail(email)) {
18        emailError.textContent = "Please enter a valid email address.";
19        isValid = false;
20    }
21
22    // Validate Phone Number (India)
23    if (!validator.isMobilePhone(phone, "en-IN")) {
24        phoneError.textContent = "Please enter a valid phone number.";
25        isValid = false;
26    }
27
28    // If valid
29    if (isValid) {
30        alert("Form submitted successfully!");
31        console.log("Email:", email);
32        console.log("Phone:", phone);
33    }
34 });

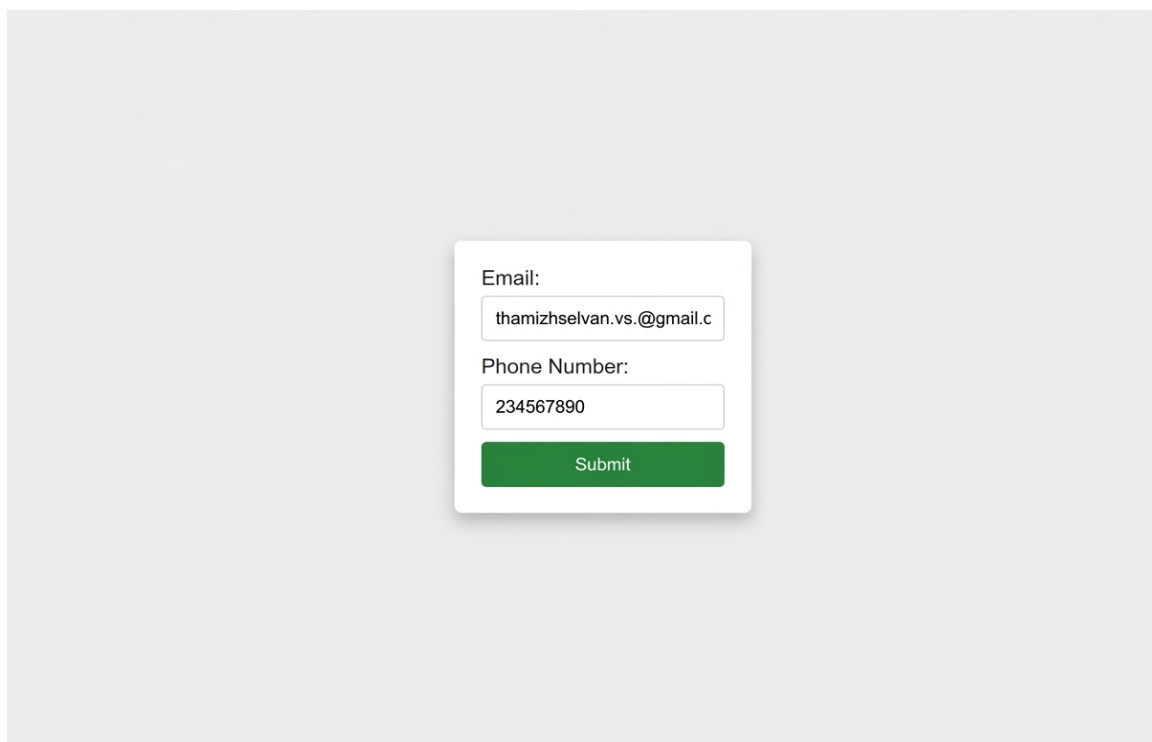
```

```

# style.css X
# style.css > body
1 body {
2     font-family: Arial, sans-serif;
3     background-color: #f4f4f4;
4     display: flex;
5     justify-content: center;
6     align-items: center;
7     height: 100vh;
8     margin: 0;
9 }
10
11 .container {
12     background-color: white;
13     padding: 20px;
14     border-radius: 5px;
15     box-shadow: 0 0 10px rgba(0, 0, 0, 0.1);
16 }
17
18 form {
19     display: flex;
20     flex-direction: column;
21 }
22
23 label {
24     margin-bottom: 5px;
25 }
26
27 input {
28     margin-bottom: 10px;
29     padding: 10px;
30     border: 1px solid #ccc;
31     border-radius: 3px;
32 }
33
34 button {
35     padding: 10px;
36     background-color: #28a745;
37     color: white;
38     border: none;
39     border-radius: 3px;
40     cursor: pointer;
41 }
42
43 button:hover {
44     background-color: #218838;
45 }
46
47 .error {
48     color: red;
49     font-size: 0.875em;
50 }

```

```
index.html X
index.html > html > body > div.container > form#myForm > input#email
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <meta charset="UTF-8">
5      <meta name="viewport" content="width=device-width, initial-scale=1.0">
6      <title>Form Validation</title>
7      <link rel="stylesheet" href="style.css">
8  </head>
9  <body>
10
11  <div class="container">
12      <form id="myForm">
13          <label for="email">Email:</label>
14          <input type="text" id="email" name="email" required>
15          <span id="emailError" class="error"></span>
16
17          <label for="phone">Phone Number:</label>
18          <input type="text" id="phone" name="phone" required>
19          <span id="phoneError" class="error"></span>
20
21          <button type="submit">Submit</button>
22      </form>
23  </div>
24
25  <!-- Validator.js CDN -->
26  <script src="https://cdnjs.cloudflare.com/ajax/libs/validator/13.6.0/validator.min.js"></script>
27
28  <script src="script.js"></script>
29
30
31 </body>
32 </html>
```



Results

The implementation successfully validates the form inputs based on the following criteria:

- **Email Validation:** Successfully identifies incorrect email formats (e.g., missing "@" or domain).
- **Phone Validation:** Specifically validates Indian phone numbers using the en-IN locale.
- **Visual Feedback:** The UI updates dynamically to show red error text when validation fails and clears previous errors upon re-submission.
- **Final Output:** As shown in the UI screenshot, a correctly filled form allows for successful "submission" simulated via a browser alert.